

ORLINGA, Russian Federation

WMO: 303280

Lat: 56.050N

Lon: 105.833E

Elev: 338

StdP: 97.33

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -43.6	(c) -40.9	(d) -45.4	(e) 0.0	(f) -43.0	(g) -42.9	(h) 0.1	(i) -40.5	(j) 5.8	(k) -17.4	(l) 5.2	(m) -16.6	(n) 0.0	(o) 250	(p) 0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.8	(c) 28.8	(d) 18.1	(e) 26.8	(f) 17.5	(g) 24.7	(h) 16.5	(i) 19.8	(j) 26.1	(k) 18.9	(l) 24.7	(m) 17.9	(n) 23.0	(o) 1.6	(p) 0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.8	13.3	21.9	16.8	12.4	21.2	15.8	11.7	20.2	58.5	26.0	55.3	24.6	52.1	23.1	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.9	(b) 4.8	(c) 4.0	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 5.9	(b) 4.8	(c) 4.0	(e) -45.1	(f) 20.9	(g) 2.2	(h) 1.5	(i) -46.7	(j) 21.9	(k) -48.0	(l) 22.8	(m) -49.2	(n) 23.6	(o) -50.8	(p) 24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.0	-26.2	-20.9	-11.9	-0.4	7.1	15.3	18.0	14.4	6.9	-1.1	-14.4	-23.4
	DBStd	16.39	8.41	7.79	6.85	4.74	3.68	3.73	2.81	3.22	3.93	5.03	9.03	9.10
	HDD10.0	5318	1122	865	680	313	107	4	0	5	109	345	733	1036
	HDD18.3	7823	1380	1098	938	563	349	104	41	126	343	603	983	1294
	CDD10.0	583	0	0	0	0	16	163	247	142	16	0	0	0
	CDD18.3	46	0	0	0	0	0	12	30	4	0	0	0	0
	CDH23.3	718	0	0	0	0	13	259	371	68	7	0	0	0
	CDH26.7	180	0	0	0	0	2	62	101	13	1	0	0	0
Wind	WSAvg	1.0	0.7	0.7	0.8	1.4	1.6	1.1	1.0	0.9	1.1	1.2	1.1	0.9
Precipitation	PrecAvg	461	21	16	15	19	32	57	79	82	46	35	29	25
	PrecMax	599	40	30	39	53	60	104	194	140	81	68	46	51
	PrecMin	330	7	6	3	5	5	10	19	45	3	5	16	10
	PrecStd	71	9	7	9	10	14	24	37	27	17	14	9	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.6	-0.6	8.1	15.4	25.7	31.0	32.0	28.1	23.2	13.3	3.6	-2.2
		MCWB	-6.3	-2.7	2.1	7.0	13.1	19.0	20.1	18.3	15.2	7.8	1.3	-3.1
	2%	DB	-8.7	-4.2	4.4	12.3	22.0	28.5	29.6	25.0	18.2	9.4	1.2	-6.1
		MCWB	-9.4	-5.7	-0.2	5.5	11.6	17.8	18.7	17.2	12.1	5.6	-0.2	-7.1
	5%	DB	-11.6	-7.3	1.4	9.8	18.8	26.5	27.4	22.8	15.6	7.0	-0.6	-8.0
		MCWB	-12.2	-8.3	-1.8	3.9	9.8	16.7	18.2	16.4	10.7	4.6	-1.8	-8.7
	10%	DB	-14.5	-9.6	-0.8	7.1	15.8	24.4	25.3	20.8	13.3	4.8	-2.6	-10.4
		MCWB	-15.1	-10.4	-3.3	2.3	8.7	15.7	17.6	15.6	9.6	2.8	-3.6	-11.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.3	-2.6	2.2	7.7	14.6	20.1	21.4	19.7	16.0	8.1	1.6	-3.2
		MCDB	-5.7	-0.7	7.2	14.1	23.3	27.4	28.3	26.5	23.2	11.4	2.6	-2.2
	2%	WB	-9.3	-5.7	0.2	5.9	12.2	19.0	20.3	18.4	13.4	6.3	-0.1	-7.0
		MCDB	-8.6	-4.4	3.7	11.4	20.4	26.3	26.4	23.3	17.0	8.8	1.2	-6.2
	5%	WB	-12.2	-8.2	-1.6	4.2	10.6	17.9	19.5	17.3	11.6	4.8	-1.8	-8.7
		MCDB	-11.6	-7.3	1.2	8.9	17.7	24.4	25.2	21.5	14.4	6.9	-0.7	-8.1
	10%	WB	-15.0	-10.4	-3.2	2.8	9.2	16.6	18.6	16.1	10.1	3.0	-3.5	-10.9
		MCDB	-14.5	-9.6	-0.6	6.4	15.0	22.3	23.9	19.6	12.9	4.5	-2.6	-10.4
Mean Daily Temperature Range	5% DB	MDBR	9.2	12.9	17.2	12.4	13.4	12.8	11.1	10.2	7.9	8.3	8.5	
		MCDBR	9.1	12.0	18.4	16.2	20.4	19.1	17.8	15.1	14.8	11.1	7.7	7.9
		MCWBR	8.7	10.7	14.1	9.5	10.7	8.9	8.1	7.5	8.4	7.3	6.4	7.4
	5% WB	MCDBR	8.7	11.8	17.8	14.7	17.9	16.5	14.5	13.2	13.3	10.4	7.3	8.0
		MCWBR	8.3	10.6	13.8	8.9	10.1	8.0	7.5	7.3	7.8	7.1	6.3	7.4
Clear-Sky Solar Irradiance	taub	0.279	0.312	0.335	0.367	0.420	0.451	0.481	0.411	0.349	0.327	0.293	0.272	
	taud	2.002	1.989	2.015	2.097	2.148	2.160	2.095	2.286	2.425	2.309	2.068	1.968	
	Ebn at Noon	581	716	807	843	818	795	758	796	807	716	559	476	
	Edh at Noon	67	101	127	136	139	139	146	113	85	74	62	54	
All-Sky Solar Radiation	RadAvg	0.83	1.81	3.19	4.44	5.16	5.68	5.24	4.33	2.82	1.50	0.79	0.55	
	RadStd	0.08	0.12	0.17	0.33	0.41	0.44	0.52	0.30	0.26	0.18	0.09	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page